

NIKITA OVRAMENKO

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EDUCATION

Toronto Metropolitan University

Bachelor of Engineering in Computer Engineering

Toronto, ON

Expected May 2027

- **Relevant coursework:** Digital Systems Engineering (in progress), Embedded Systems Design (in progress), Computer Organization and Architecture, Microprocessor Systems.

TECHNICAL SKILLS

Languages: C, C++, Embedded C, VHDL, HCS12 Assembly, Python, Java, TypeScript, SQL

Hardware/Embedded: FPGA design, digital logic, computer architecture, ARM Cortex-M3, memory-mapped I/O, interrupts, ADC, state machines

Frameworks/Software: Angular, RxJS, React, Spring Boot, Node.js/Express, GraphQL, PostgreSQL

Tools/Systems: Quartus II, Xilinx CAD tools, CodeWarrior, Git, Linux, Docker, AWS, Home Assistant, MQTT

EXPERIENCE

Software Engineering Intern — International Farming

FarmAdvisor Agricultural Technology Platform

Raleigh, NC

Apr 2026 – Aug 2026

- Implemented a TCP-inspired adaptive batching algorithm in **Angular**, **TypeScript**, and **GraphQL** that reduced request size after failures, increased it after successful transfers, and retained the largest reliable batch size.
- Redesigned a two-page upload and table-creation flow as a real-time **RxJS** wizard, synchronizing AWS upload progress with pending, success, and failure states.
- Extended a shared Angular table used across **22+ application views** and built a lazy-loaded GraphQL hierarchy supporting nested folders, multi-select, and drag-and-drop movement.

Full-Stack Developer — Smart Home Client Project

React PWA / Home Assistant / Networked Embedded Systems

Toronto, ON

May 2025 – Aug 2025

- Designed and delivered a paid smart-home system for **10–15 residents**, integrating a wall-mounted React/TypeScript dashboard with Home Assistant and networked IoT hardware.
- Integrated **40+ Zigbee, Z-Wave, and MQTT devices** across lighting, covers, climate, media, sensors, and access control; developed Python automations and connectivity tests.
- Configured Tailscale VPN and Asterisk/DoorBird SIP services for secure remote access and two-way doorbell calling; diagnosed device and network failures using Linux tools and system logs.

IoT & Home Automation Developer Intern — Pine Smart Solutions

Home Assistant / Networking / IoT Systems

Toronto, ON

Jan 2025 – Apr 2025

- Configured, validated, and troubleshooted residential automations using Home Assistant, MQTT, Python, and Ubiquiti networking equipment.

HARDWARE & EMBEDDED PROJECTS

32-bit Semi-RISC Processor | VHDL, Quartus II, Cyclone IV FPGA

- Built and simulated a multi-cycle, Harvard-style load/store processor by integrating a program counter and register set, 32-bit ALU, 1 KB data memory, datapath, hardwired control unit, and reset circuitry.
- Verified fetch, pre-decode, and execute behavior across arithmetic, logic, memory, branch, and jump instructions using timing simulations; synthesized and tested the ALU on a DE2-115 FPGA.

Autonomous eebot Firmware | HCS12 Assembly, CodeWarrior

- Programmed HCS12 firmware for autonomous navigation using a state-machine dispatcher, five analog sensors, ADC acquisition, memory-mapped I/O, motor and bumper control, and LCD telemetry.
- Implemented interrupt-driven timing and collision recovery, then calibrated sensor thresholds and modularized movement, sensing, and display routines to simplify integration and debugging.